Project 95: Integrated Manufacturing Production Planner

Python Mini Project | College Assignment

1. Aim

To develop a basic Python program that checks raw material availability, production feasibility, product defects, and customer order requirements.

2. Algorithm

1. Start the program.
2. Store product names, raw material requirements, and available stock.
3. Store production quantities, defect counts, and customer order demand.
4. Calculate total wood and steel requirements.
5. Compare requirements with stock and identify shortages.
6. Calculate good products after excluding defective items.
7. Compare good products with customer order demand.
8. Calculate defect percentages and check quality standards.
9. Display the complete production report.
10. Stop.

3. Flowchart (Text Representation)

```
START
  |
Store product, material, production, defect and order data
  |
Calculate raw material requirements
  |
Check available stock -> Identify shortages if any
  |
Calculate good products and defect percentage
  |
Check customer orders and quality limit
  |
Display production and quality report
  |
STOP
```

4. Complete Python Program

```
# Project 95: Integrated Manufacturing Production Planner

print("=====================================")
print("  MANUFACTURING PRODUCTION PLANNER")
print("=====================================")

# Product details
products = ["Chairs", "Tables", "Desks"]

# Raw materials required per product (Wood, Steel)
materials = [(3, 2), (5, 4), (4, 3)]
# Available raw materials
```

```
available_wood = 100
available_steel = 80

# Production details
production = [10, 8, 6]
defects = [1, 0, 1]
orders = [9, 10, 5]

# Maximum allowed defect percentage
quality_limit = 10

total_wood = 0
total_steel = 0

print("\nPRODUCT DETAILS")

for i in range(len(products)):
    wood, steel = materials[i]
    wood_needed = wood * production[i]
    steel_needed = steel * production[i]
    total_wood += wood_needed
    total_steel += steel_needed

    print("\nProduct:", products[i])
    print("Production:", production[i])
    print("Wood Required:", wood_needed)
    print("Steel Required:", steel_needed)

print("\n======================================")
print("RAW MATERIAL REPORT")
print("======================================")
print("Available Wood:", available_wood)
print("Required Wood:", total_wood)
print("Available Steel:", available_steel)
print("Required Steel:", total_steel)

wood_shortage = 0
steel_shortage = 0

if total_wood > available_wood:
    wood_shortage = total_wood - available_wood

if total_steel > available_steel:
    steel_shortage = total_steel - available_steel

if wood_shortage == 0 and steel_shortage == 0:
    print("Production Status: FEASIBLE")
else:
    print("Production Status: NOT FEASIBLE")
    print("Wood Shortage:", wood_shortage)
    print("Steel Shortage:", steel_shortage)

print("\n======================================")
print("QUALITY AND ORDER REPORT")
print("======================================")

for i in range(len(products)):
    good_products = production[i] - defects[i]
    defect_percentage = defects[i] / production[i] * 100

    print("\nProduct:", products[i])
    print("Produced:", production[i])
    print("Defective:", defects[i])
    print("Good Products:", good_products)
    print("Order Demand:", orders[i])
    if good_products >= orders[i]:
```

```
        print("Order Status: Fulfilled")
    else:
        shortage = orders[i] - good_products
        print("Order Status: Shortage")
        print("Units Short:", shortage)

    if defect_percentage <= quality_limit:
        print("Quality Status: PASS")
    else:
        print("Quality Status: VIOLATION")

    print("Defect Percentage:", round(defect_percentage, 2), "%")

print("\n====================================")
print("       REPORT COMPLETED")
print("====================================")
```

5. Sample Output

```
====================================
  MANUFACTURING PRODUCTION PLANNER
====================================

PRODUCT DETAILS

Product: Chairs
Production: 10
Wood Required: 30
Steel Required: 20

Product: Tables
Production: 8
Wood Required: 40
Steel Required: 32

Product: Desks
Production: 6
Wood Required: 24
Steel Required: 18

====================================
RAW MATERIAL REPORT
====================================
Available Wood: 100
Required Wood: 94
Available Steel: 80
Required Steel: 70
Production Status: FEASIBLE

====================================
QUALITY AND ORDER REPORT
====================================

Product: Chairs
Produced: 10
Defective: 1
Good Products: 9
Order Demand: 9
Order Status: Fulfilled
Quality Status: PASS
Defect Percentage: 10.0 %

Product: Tables
Produced: 8
```

```
Defective: 0
Good Products: 8
Order Demand: 10
Order Status: Shortage
Units Short: 2
Quality Status: PASS
Defect Percentage: 0.0 %

Product: Desks
Produced: 6
Defective: 1
Good Products: 5
Order Demand: 5
Order Status: Fulfilled
Quality Status: VIOLATION
Defect Percentage: 16.67 %

====================================
       REPORT COMPLETED
====================================
```

6. Python Concepts Used

Syllabus Topic	Application
Variables and data types	Store stock and production values
Operators	Calculate requirements and defect percentages
Conditional statements	Check feasibility, shortages, and quality
Loops	Process multiple products
Strings	Product names and report messages
Lists	Production quantities, defects, and orders
Tuples	Material quantities per product
Tuple unpacking	wood, steel = materials[i]

7. Result

The Integrated Manufacturing Production Planner was developed using basic Python concepts. In the sample data, the planned production is feasible because the raw materials are sufficient. However, tables have an order shortage of 2 units and desks exceed the 10% defect limit.

Note: This is a basic syllabus-level solution using fixed sample data. Change the values in the lists to test other manufacturing situations.